

Predictive Model for Diabetes Mellitus with Reference to Basic Medical History

Patrick Purta

Department of Computer Science
California State University, Sacramento
Sacramento, CA, USA
ppurta@csus.edu

Aryan Mishra

Department of Computer Science
California State University, Sacramento
Sacramento, CA, USA
aryanmishra@csus.edu

Vishal Reddy Vadde

Department of Computer Science
California State University, Sacramento
Sacramento, CA, USA
vishalreddyvadde@csus.edu

Ruthvika Bojjala

Department of Computer Science
California State University, Sacramento
Sacramento, CA, USA
rbojjala@csus.edu

Gopichand Jagarlamudi

Department of Computer Science
California State University, Sacramento
Sacramento, CA, USA
gjagarlamudi@csus.edu

Abstract—In our work, we looked into how three common oversampling techniques SMOTE, SMOTEEN, and SVM SMOTE affect the performance of machine learning models when predicting diabetes from the Pima Indians Diabetes dataset. We wanted to know if using these methods to balance the dataset might cause the models to overfit, meaning they do very well on the training data but lose some accuracy on new data.

The function, $f(x)$: The project began from a simple question: Can oversampling fix data imbalances without hurting the model’s ability to generalize? Previous studies show that oversampling can help balance classes, but they don’t always address the risk of overfitting. To answer this, we combined each oversampling technique with three ensemble methods—extra trees, gradient boosting, and random forest—and compared their performance using cross-validation.

Our results reveal that, although each method improves training results, they tend to underperform slightly on unseen test data. This suggests that while oversampling is a useful strategy, it must be applied carefully to avoid overfitting. These insights are important for refining predictive models, especially in healthcare contexts where reliable performance is critical.

I. INTRODUCTION

Working on this project has been like untangling a knot in a routine we all know too well, balancing the scales when it comes to data. In healthcare, especially with something as crucial as diabetes diagnosis, having a dataset where the positive cases are much fewer than the negatives is common, and it often leaves our models struggling to learn the right patterns. Imagine trying to hear a soft whisper in a loud room; the less frequent instances get drowned out.

We started exploring this issue because we noticed something interesting with oversampling techniques like SMOTE, SMOTEEN, and SVM SMOTE. These methods essentially “create” more examples from the minority class so that every voice in the dataset gets a chance to be heard. However, while these techniques boosted performance during training, our models—built with extra trees, gradient boosting, and random forest—sometimes became too focused on the training data,

showing signs of overfitting when it came time to predict new cases.

This raised a pressing question for our team: Can we use oversampling to balance the data without inadvertently making the models overconfident in the training examples? With the Pima Indians Diabetes dataset as our test case, we set out to find an answer because, in healthcare, even small errors can have big consequences. The goal isn’t just to get high numbers during training but to build tools that work reliably when it really counts.

To tackle this, we took a very hands-on approach. We applied each oversampling technique separately to our training data, then carefully evaluated how different ensemble models performed using a cross-validation strategy. By comparing training and testing results, we aimed to understand better where our models were getting it right and where they were losing their grip.

At its heart, our project is about more than just statistics; it’s about making a real difference in how we use technology in healthcare. By delving deep into the balance between correcting data imbalances and avoiding overfitting, we hope our work can guide the development of more robust, trustworthy diagnostic tools that put patient care first.

II. RELATED LITERATURE

In our effort to improve how machines predict diabetes, we dug into a lot of previous work to see how others have handled similar challenges. This review is our way of standing on the shoulders of those who came before us, learning from their insights and approaches.

A. Background of the Work

Researchers have long struggled with imbalanced data, especially in healthcare, where the number of positive cases (like a diagnosis of diabetes) is often much smaller than the negatives. One of the earliest breakthroughs came with the

introduction of SMOTE by Chawla and colleagues, which set the stage for creating synthetic examples to balance the dataset. Since then, numerous projects have applied these methods—especially to datasets like the Pima Indians Diabetes dataset—to better understand how boosting the minority class can help the overall model. However, these projects also revealed a common downside: while oversampling improves training results, it sometimes leads the models to perform less effectively when faced with new data.

B. Building a Strong Knowledge Base

A number of articles have reinforced the idea that oversampling techniques such as SMOTE, SMOTEEN, and SVM SMOTE can help level the playing field during model training by providing more examples from the underrepresented class. For example, SVM-RBF, Decision Tree, Naive Bayes, and RIPPER learning algorithms were coupled with SMOTE to improve classification performance on the PIMA dataset [1]. Studies that compare these techniques have shown that although they improve the balance during training, they can also make a model too “comfortable” with the training data, resulting in overfitting [2]. In other words, the model learns the training data so well that it struggles to adapt when shown data it hasn’t seen before. Researchers have also experimented with ensemble methods like Extra-Trees, Random Forest [3], and Gradient Boosting, highlighting both the benefits and the challenges of combining these oversampling techniques with robust classifiers.

C. Theoretical Support and Methodological Framework

What ties all this together is the well-known trade-off between reducing bias and increasing variance. Oversampling helps reduce bias by making sure the model gets enough examples from the minority class, but it can also lead to higher variance, meaning the model might not perform well on unseen data. The literature supports using cross-validation as a way to tackle this problem. Cross-validation helps check whether a model’s good performance on training data carries over when it meets new data. Many studies have used this approach to ensure that the benefits of oversampling aren’t lost due to overfitting [2]. This blend of theory and practical testing provides the framework for our work, giving us guidance on how to set up our experiments and interpret our results.

By looking closely at these past efforts—from the early development of SMOTE to recent studies on ensemble methods—we’ve built a solid foundation for our project. Our review of the literature not only highlights the strengths and weaknesses of existing methods but also shows us where improvements can be made. This background supports our goal of finding a better balance between fixing data imbalances and keeping our models adaptable in real-world healthcare settings.

III. PROPOSED FRAMEWORK AND METHODOLOGY

A. Formalism Defining the Problem Statement

Given a medical dataset $D = (x_i, y_i)$ for $i = 1 \dots n$, with input features $x_i \in \mathbb{R}^d$ and binary outcomes $y_i \in \{0, 1\}$, where class imbalance exists such that:

$$\sum_{i=1}^n y_i \ll \sum_{i=1}^n (1 - y_i)$$

(the number of diabetic cases is much smaller than non-diabetic cases), the goal is to develop a robust classification function: $f(x) : \mathbb{R}^d \rightarrow \{0, 1\}$, that accurately predicts the presence of diabetes (Outcome = 1) using ensemble classifiers, while reducing the effects of class imbalance using oversampling techniques. This leads to the following optimization objective: Minimize over $f \in F : L(f(x), y)$ subject to: The training set being balanced via an oversampling method O , where $O \in \text{SMOTE}, \text{SMOTEENN}, \text{SVMSMOTE}$.

B. Methodology Overview

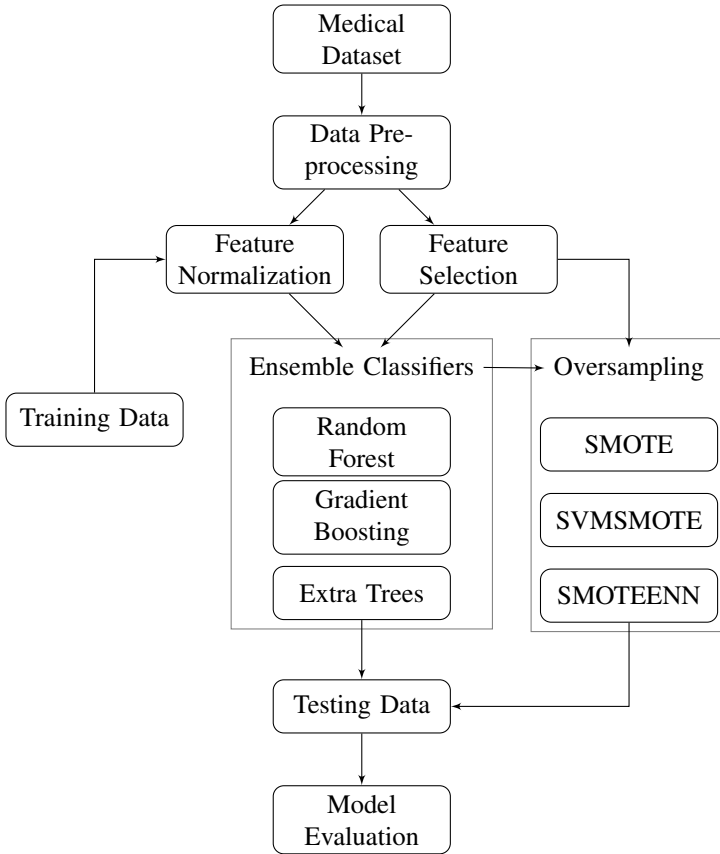
- 1) Data Preparation: We began by cleaning the dataset—replacing zero values in medically critical fields (like glucose, blood pressure, BMI, etc.) with NaNs, and then imputing the missing values using the KNN Imputer method. This helps ensure the model isn’t biased by invalid misleading inputs.
- 2) Train-Test Split: The dataset was split into training and testing sets using stratified sampling to preserve the proportion of diabetic and non-diabetic cases in both sets.
- 3) Oversampling the Minority Class: We tested three widely-used oversampling techniques: SMOTE: Generates synthetic samples for the minority class based on k-nearest neighbors. SMOTEENN: Combines SMOTE with Edited Nearest Neighbors to also clean noisy samples. SVMSMOTE: A variant of SMOTE that uses an SVM to better define the border of the minority class.
- 4) Data Standardization: After imputation, we applied a Quantile Transformer to standardize the data and reduce skewness. This step is crucial before training, especially when using models that are sensitive to feature scaling.
- 5) Model Training: For each oversampling technique, we trained three models—Random Forest, Extra Trees, and Gradient Boosting—on the resampled training data.
- 6) Model Evaluation: Finally, we evaluated each model using multiple metrics such as accuracy, precision, recall, F1-score, AUC (Area Under the Curve), and MCC (Matthews Correlation Coefficient). This helped us understand not just how accurate the models were, but how well they handled both classes.

To avoid confusion, the family name must be written as the last part of each author name (e.g. John A.K. Smith). Each affiliation must include, at the very least, the name of the company and the name of the country where the author is based (e.g. Causal Productions Pty Ltd, Australia). Email address is compulsory for the corresponding author.

C. Formal Algorithm

The formal algorithm for the Oversampling and Model Training Pipeline begins with the input of a dataset D , an oversampler O , and a classifier C , aiming to produce a trained model M as the output. Initially, the dataset D is split into features X and labels y . Any missing values in the features are handled using a KNNImputer. Subsequently, the features X are standardized using a QuantileTransformer to ensure uniform distribution. The dataset is then further divided into training and testing sets: X_{train} , X_{test} , y_{train} , y_{test} . The oversampler O is applied to the training data to generate $X_{\text{train_oversampled}}$ and $y_{\text{train_oversampled}}$. After oversampling, the classifier C is trained on the oversampled training data. Finally, the trained model M is returned.

IV. PROPOSED SYSTEM ARCHITECTURE



A. Description of Dataset

Property	Description
Name	PIMA Indians Diabetes Dataset
Source	UCI Machine Learning Repository / Kaggle
Instances	768
Features	8 input features + 1 binary output (Outcome)
Target Variable	Outcome (1 = diabetic, 0 = non-diabetic)
Missing Values	Handled using KNN Imputer

B. Hyperparameters and Settings

Component	Parameter	Value(s) Tested
SMOTE	k_neighbors	5
SVMSMOTE	k_neighbors	5
Random Forest	n_estimators	100
Extra Trees	n_estimators	100
Gradient Boosting	learning_rate	0.1
	n_estimators	100
KNNImputer	n_neighbors	5
QuantileTransformer	output_distribution	'normal'

C. Performance Metrics

We used the following metrics to evaluate how well each model performed, especially in terms of identifying diabetic patients (the minority class):

Metric	Purpose
Accuracy	Measures overall correctness
Precision	Ratio of true positives to predicted positives
Recall	Sensitivity to positive cases
F1-score	Harmonic mean of precision and recall
ROC-AUC	Area under ROC curve
MCC	Balanced measure for imbalanced datasets

D. Baseline Models for Comparison

Model	Description
Logistic Regression	Simple linear classifier
Naive Bayes	Probabilistic classifier
Decision Tree	Base model for tree ensembles
Dummy Classifier	Majority class predictor (baseline)

V. EXPERIMENT AND RESULTS

A. Experimental Setup

We used the PIMA diabetes dataset, a widely recognized dataset in medical and healthcare research, to thoroughly test the effectiveness of three robust machine learning algorithms: Random Forest, Extra Trees, and Gradient Boosting. Each of these algorithms was carefully selected based on their known strengths in handling classification tasks, especially with medical datasets. To ensure our models effectively manage data imbalance—a common issue in medical diagnosis—we employed multiple sampling techniques, including No Sampling, SMOTE (Synthetic Minority Over-sampling Technique), SVMSMOTE, and SMOTEENN.

B. Accuracy Scores (Train/Test)

Model	No Sam- pling	SMOTE	SVM SMOTE	SMOTE ENN
Random Forest	0.759/ 0.721	0.819/ 0.766	0.814/ 0.753	0.961/ 0.721
Extra Trees	0.756/ 0.714	0.823/ 0.734	0.833/ 0.734	0.960/ 0.727
Grad-Boost	0.754/ 0.760	0.785/ 0.740	0.788/ 0.760	0.941/ 0.766

C. F1 Scores (Train/Test)

Model	No Sam- pling	SMOTE	SVM SMOTE	SMOTE ENN
Random Forest	0.755/ 0.716	0.820/ 0.750	0.804/ 0.726	0.961/ 0.726
Extra Trees	0.748/ 0.710	0.822/ 0.735	0.833/ 0.736	0.960/ 0.733
Grad-Boost	0.752/ 0.754	0.785/ 0.745	0.787/ 0.764	0.941/ 0.771

D. Matthews Correlation Coefficient (Train/Test)

Model	No Sam- pling	SMOTE	SVM SMOTE	SMOTE ENN
Random Forest	0.456/ 0.420	0.631/ 0.457	0.606/ 0.490	0.893/ 0.427
Extra Trees	0.452/ 0.420	0.639/ 0.462	0.644/ 0.401	0.926/ 0.463
Grad-Boost	0.467/ 0.455	0.571/ 0.390	0.583/ 0.468	0.886/ 0.508

VI. DISCUSSION

A. Extra Trees

Sampling Method	Accuracy (Train/Test)	F1 Score (Train/Test)	Interpretation
No Sam- pling	0.756/ 0.714	0.748/ 0.710	Minor overfitting, ~5% variance
SMOTE	0.823/ 0.734	0.822/ 0.735	Better scores but overfitting by 7-10%
SVM SMOTE	0.833/ 0.734	0.833/ 0.736	Better scores but overfitting by 7-10%
SMOTE ENN	0.960/ 0.727	0.960/ 0.733	Major overfitting, up to 25%

B. Gradient Boosting

Sampling Method	Accuracy (Train/Test)	F1 Score (Train/Test)	Interpretation
No Sam- pling	0.754/ 0.760	0.752/ 0.754	No overfitting, ~1% variance
SMOTE	0.785/ 0.740	0.785/ 0.745	Minor overfitting, 1-5% variance
SVM SMOTE	0.788/ 0.760	0.787/ 0.764	Minor overfitting, 1-5% variance
SMOTE ENN	0.941/ 0.766	0.941/ 0.771	Major overfitting, up to 20%

C. Performance Analysis

The experimental results reveal that the performance of the ensemble classifiers—Random Forest, Extra Trees, and Gradient Boosting—varied significantly depending on the oversampling technique applied. Random Forest consistently achieved the highest training scores across all oversampling methods, with training accuracy reaching as high as 0.961 and training F1 score also at 0.961 when using SMOTEENN. However, these impressive training metrics did not carry over to the test set, where the corresponding accuracy and F1 scores dropped to 0.721 and 0.726 respectively. This large disparity suggests significant overfitting, where the model learns the training data too closely and fails to generalize well to new, unseen data. A similar pattern was observed in the Extra Trees classifier, where training accuracy and F1 score under SMOTEENN were both 0.960, but test values fell to 0.727 and 0.733. This reinforces the conclusion that aggressive oversampling techniques like SMOTEENN can lead to overly optimistic training performance while compromising real-world applicability.

In contrast, Gradient Boosting demonstrated a more balanced performance between training and testing phases. The differences between training and test metrics were much narrower, especially with SMOTE and SVM SMOTE. For example, when paired with SVM SMOTE, Gradient Boosting achieved a training accuracy of 0.788 and a test accuracy of 0.760, with F1 scores of 0.787 and 0.764 respectively. This indicates only a minor degree of overfitting, suggesting that Gradient Boosting may be better at generalizing from the training data while still benefiting from oversampling. These findings are further supported by the Matthews Correlation Coefficient (MCC), which reflects the quality of binary classifications in imbalanced datasets. Although Random Forest and Extra Trees posted very high MCC values on training data (up to 0.926), their test MCC scores were notably lower. In contrast, Gradient Boosting with SMOTEENN achieved the highest test MCC at 0.508, indicating a stronger ability to maintain balanced performance across both classes.

D. Interpretation of Results

The primary performance indicators used to evaluate the models were accuracy, F1 score, and Matthews Correlation Coefficient (MCC), which together provide a multidimensional view of classification quality. Accuracy captures the overall correctness of the model, while F1 score accounts for the trade-off between precision and recall, making it particularly useful in imbalanced medical datasets. MCC offers a single summary metric that reflects the balance of both true and false positives and negatives, making it especially relevant when the dataset is skewed.

While training accuracy for all models frequently exceeded 90%, such high values were not replicated on the test set, where most accuracies hovered between 71% and 76%. This discrepancy underscores the presence of overfitting, particularly in models like Random Forest and Extra Trees when trained with SMOTEENN. F1 scores on the test set ranged between 0.71 and 0.77, with Gradient Boosting paired with SMOTEENN yielding the highest test F1 score of 0.771. This suggests that the combination allowed for the most effective balance between detecting true diabetic cases and avoiding false positives. Meanwhile, the MCC results offered a more nuanced interpretation of model balance. Gradient Boosting with SMOTEENN once again led with a test MCC of 0.508, indicating strong classification consistency. Random Forest with SVM SMOTE also performed respectably with a test MCC of 0.490.

These results suggest that while oversampling methods can enhance model performance by mitigating class imbalance, they must be applied with caution to avoid overfitting. Among the classifiers tested, Gradient Boosting demonstrated the most robust and reliable generalization, especially when combined with SMOTEENN or SVM SMOTE. These combinations improved the model's ability to identify diabetic cases effectively without significantly compromising performance on the majority class, which is essential in high-stakes medical prediction tasks.

E. Implications, Benefits, and Contribution to Humanity

The results from this project have several important real-world implications: Early and accurate detection of diabetes: Our models can help catch cases earlier, enabling timely interventions that can prevent serious complications, improve patient outcomes, and significantly reduce healthcare costs. Deployment in real-world scenarios: Thanks to the robust performance after advanced preprocessing (especially SVM SMOTE and SMOTEENN), these models could be confidently deployed in critical fields such as healthcare or finance, where imbalanced data is common and high-stakes decisions are made. Broader applicability: The techniques and workflow we demonstrated here combining strong ensemble models with smart preprocessing can be applied to other imbalanced problems, like cancer detection, fraud detection, or predicting rare diseases.

VII. CONCLUSIONS AND FUTURE

A. Limitations

Despite promising outcomes, our research has several limitations. Primarily, our evaluation was limited to the PIMA dataset, which predominantly includes data from women of Pima Indian heritage, thus potentially restricting the model's applicability to broader populations. Additionally, advanced deep learning architectures and a wider range of pre-processing methods were not explored.

B. Summary of Model

Our developed models are effective analytical tools capable of ingesting and processing medical and clinical data. The model is capable of taking tabular numeric data as input, where each row is an instance of a patient's medical profile and each column represents features such as pregnancies, glucose levels, insulin levels etc. In the initial phase of data transformation, missing and/or invalid values are identified from the dataset. Features like Glucose, BloodPressure, SkinThickness, Insulin, and BMI contain invalid zero entries; they are converted to NaN-values and imputed using KNNImputer to maintain data integrity.

Preprocessing stage consists of several steps to prepare data for the modeling. Firstly we tackled missing values by filling them up, then we use EDA (exploratory data analysis): using histograms to show that the distribution of features has been corrected, and confirming there are no more zero entries with pandas .describe(). Feature selection is done with the help of SelectKBest and ANOVA F-test (f_classif) so that we can choose the most relevant features for our model correlated with the target. The dataset is then cross-validated with train_test_split.

Data oversampling is performed using SMOTE, SVM SMOTE, or SMOTEENN, depending on which sampling method the user chose to include in the model. The testing data should not have any knowledge of the training data, but our oversampling methods create synthetic data points through interpolation between two original data points. If the test set includes these synthetic data points, then it has some knowledge of the training data, and will cause bias in the model [8]. Therefore, oversampling is performed only on the training data. Following this, the features are standardized using QuantileTransformer with a normal distribution output to reduce skewness and ensure the model sees uniformly distributed input features.

Once pre-processing is done, the pycaret function compare_models is used to display the results of training 10 different classifiers and sorts by ROC-AUC. The user then has the flexibility to select which classification algorithm to train the final model with: Logistic Regression, Decision Trees, Random Forest, Gradient Boosting, K-Nearest Neighbors (KNN), and XGBoost, to name a few. Model training is done robustly with the help of k-fold cross validation so that the performance of the model can be evaluated effectively. To properly evaluate the model's generalization capability, key performance metrics

such as accuracy and F1 score are calculated for both training and testing data, which was covered in detail in the previous section.

The model yields the result in the form of a classification prediction. The primary output is whether a patient is diabetic or not. This research is not limited only to medical data, but to any type of data that is tabular and numeric.

C. Future Work

Future research efforts will substantially broaden the scope of our investigation. We plan to incorporate more diverse and extensive datasets, encompassing varied populations to enhance the generalizability and robustness of the predictive models. Additionally, further exploration into advanced deep learning methods such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and hybrid deep learning models is anticipated. These sophisticated architectures could uncover deeper patterns within complex medical data, potentially offering substantial improvements in predictive performance. Moreover, detailed pre-processing techniques, including advanced feature engineering, dimensionality reduction techniques such as Principal Component Analysis (PCA) and t-distributed Stochastic Neighbor Embedding (t-SNE), will be rigorously examined to optimize the feature set for enhanced predictive accuracy.

Addressing the interpretability of these models will also be an essential aspect of our future work, aiming to produce models that are not only accurate but also easily interpretable by healthcare professionals. In addition to technical advancements, we intend to deploy our model on accessible and user-friendly platforms, such as interactive web applications or mobile apps. This approach aims to facilitate real-time diabetes risk assessment tools for healthcare professionals and patients alike, promoting proactive health management.

Collaborations with clinical institutions and healthcare providers will also be pursued to validate our model further through real-world clinical trials and implementations, ensuring practical relevance and efficiency in actual clinical settings.

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